

PHYSICS SYLLABUS - Class 12th, JKBOSE

Max Marks: 100
Theory: 70 Marks

Time Allowed: 3 hrs
Practical: 30 Marks

Unit 1: Electrostatics (09 Marks)

Electric Charges and Fields

- Electric charges, conservation of charge
- Coulomb's law
- Force between two point charges
- Forces between multiple charges
- Superposition principle
- Continuous charge distribution
- Electric field and electric field lines
- Electric field due to a point charge
- Electric dipole
- Electric field due to a dipole
- Torque on a dipole in uniform electric field
- Electric flux
- Gauss's theorem and applications
- Field due to infinitely long straight wire
- Uniformly charged infinite plane sheet
- Uniformly charged thin spherical shell

Electrostatic Potential and Capacitance

- Electric potential and potential difference

- Potential due to point charge and dipole
- Equipotential surfaces
- Electrical potential energy of system of charges
- Conductors and insulators
- Free and bound charges inside conductor
- Dielectrics and polarization
- Capacitors and capacitance
- Combination of capacitors (series and parallel)
- Parallel plate capacitor with and without dielectric
- Energy stored in capacitor (formula only)

Unit 2: Current Electricity (07 Marks)

- Electric current
- Flow of electric charges in a metallic conductor
- Drift velocity and mobility
- Relation with electric current
- Ohm's law
- V-I characteristics
- Electrical energy and power
- Electrical resistivity and conductivity
- Temperature dependence of resistance
- Internal resistance of a cell
- Potential difference and emf
- Kirchhoff's rules
- Wheatstone bridge

Unit 3: Magnetic Effects of Current and Magnetism (09 Marks)

Moving Charges and Magnetism

- Magnetic field concept
- Oersted's experiment
- Biot-Savart law
- Magnetic field due to circular current loop
- Ampere's law
- Magnetic field due to infinitely long straight wire
- Straight solenoid (qualitative treatment)
- Force on moving charge in magnetic field
- Force on current carrying conductor
- Force between parallel conductors
- Definition of ampere
- Torque on current loop
- Current loop as magnetic dipole
- Moving coil galvanometer
- Conversion to ammeter and voltmeter

Magnetism and Matter

- Bar magnet and equivalent solenoid
- Magnetic field of dipole
- Torque on magnetic dipole
- Magnetic field lines
- Magnetic properties of materials
- Diamagnetic, paramagnetic and ferromagnetic substances
- Magnetization and effect of temperature

Unit 4: Electromagnetic Induction and Alternating Current (09 Marks)

Electromagnetic Induction

- Faraday's laws
- Induced emf and current
- Lenz's law
- Self induction
- Mutual induction

Alternating Current

- Alternating current
- Peak and RMS value of AC
- Reactance and impedance
- LCR series circuit
- Resonance
- Power in AC circuits
- Power factor
- AC generator
- Transformer

Unit 5: Electromagnetic Waves (03 Marks)

- Displacement current
- Electromagnetic waves
- Properties of EM waves
- Transverse nature
- Electromagnetic spectrum
- Radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays

Unit 6: Optics (14 Marks)

Ray Optics and Optical Instruments

- Reflection of light
- Spherical mirrors
- Mirror formula
- Refraction of light
- Total internal reflection
- Optical fibres
- Refraction at spherical surfaces
- Lenses
- Thin lens formula
- Lens maker's formula
- Magnification
- Power of lens
- Combination of thin lenses
- Refraction through prism
- Optical instruments: microscopes and telescopes

Wave Optics

- Wavefront and Huygens principle
- Reflection and refraction using wavefronts
- Proof of laws of reflection and refraction
- Interference
- Young's double slit experiment
- Fringe width
- Coherent sources
- Diffraction
- Single slit diffraction
- Width of central maxima (qualitative)

Unit 7: Dual Nature of Radiation and Matter (04 Marks)

- Photoelectric effect
- Hertz and Lenard observations
- Einstein's photoelectric equation
- Particle nature of light
- Experimental study of photoelectric effect
- Matter waves
- de Broglie relation

Unit 8: Atoms and Nuclei (08 Marks)

Atoms

- Alpha particle scattering experiment
- Rutherford model
- Bohr model of hydrogen atom
- Radius of nth orbit
- Velocity and energy of electron
- Hydrogen line spectra

Nuclei

- Composition of nucleus
- Nuclear force
- Mass-energy relation
- Mass defect
- Binding energy
- Binding energy per nucleon
- Nuclear fission
- Nuclear fusion

Unit 9: Electronic Devices (07 Marks)

Semiconductor Electronics

- Energy bands in conductors, semiconductors and insulators
- Intrinsic and extrinsic semiconductors
- p-n junction diode
- I-V characteristics of diode
- Forward and reverse bias
- Diode as rectifier